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SLS-00886 Air Communication Protocol

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CONTENTS

i.	Document Notes.....	3
ii.	Version History	3
iii.	Copyright	4
iv.	Terms Explanation.....	4
1	General Notes.....	5
2	GTD Communication Protocol	6
2.1	Communication Packet	6
2.2	Descriptions	6
3	DTG Communication Protocol	9
3.1	Communication Packet	9
3.2	Descriptions	9
4	Parameter Descriptions.....	13
4.1	Basic-Parameters of DTU (346 bytes)	13
4.2	GEO-FENCING Parameters (256 bytes)	16
4.3	SMS Parameters (842 bytes)	19
5	About RITI Technology Inc.....	21

i. Document Notes

This protocol describes the communication between SLS-00886 GPS Tracker and the PC-end server.

ii. Version History

Date	Version	Description of change	Author
2012/01/20	V1.00	Origin by SLS communication protocol	Terry
2012/01/30	V1.01	1. Modify 3.2.5 25V to 12V 2. Modify 3.2.6 SEQ to VAL 3. Modify 4.1 description of FSL	Terry
2012/03/05	V1.02	1. 3.2.6 add TOW_ALERT flag 2. Modify 4.1 description of LNG 3. Modify 4.3 description of MSL	Terry
2012/03/07	V1.03	1. 3.2.6 add VOLT_ALERT flag 2. Modify 3.2.6 description of BYTE0	Terry
2012/05/16	V1.04	1. 3.2.3 modify DTG to 05 _h 2. 3.2.5 modify ADC up to 60VDC (EA60 _h) 3. 4.1 modify DTU ADC, OPx, ANx contents	Terry
2013/02/20	V1.05	1. 2.2.4 Add new 67h in GTD command 2. 3.2.4 Add new 70h in DTG command 3. 4.1 Add new FSL bits descriptions 4. Modify DTG DATA length from 907 bytes to 914 bytes 5. Firmware Compatibility: 00886-12L06-xx or above	Terry

iii. Copyright

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iv. Terms Explanation

GCP	GPRS Communication Program, installed on PC end.
DTU	Data Terminal Unit, moving tracking unit, installed on vehicles or other place. Here is referred to SLS-00886 tracking unit.
GTD	Data transmitted from GCP to DTU.
DTG	Data transmitted from DTU to GCP.
CID	Caller Identification
SYNC	Synchronization flag
THB	Terminal Heart Beat, sent by DTU in specified interval to prevent DTU from being disconnected in GPRS network.

1 General Notes

- 1.1 DTU sets up a TCP/UDP connection with GCP via a GPRS network.
- 1.2 When DTU is in automatic GPRS connection mode, it will build up a connection to GCP as long as ACC is ON. If the connection fails, DTU will try to connect repeatedly until a successful connection.
- 1.3 When DTU is in manual GPRS connection mode, it can only be set up by a SMS command.
- 1.4 After DTU connects to GCP, it will send the data corresponding to the data uploading interval set by parameters (UT1, UT2).
- 1.5 DTU will monitor the GPRS network connection and it will re-connect automatically when the current connection fails. If there is no data uploading in THB interval, DTU will send a heart beat data to keep the GPRS connection.
- 1.6 If there is no feedback from GSM module within 30 seconds after sending DTG command, DTU will save the data to flash.
- 1.7 In the first connection or re-connection, if the first set of IP/PORT is invalid DTU will try the second set of IP/PORT automatically. If both sets of IP/PORT are invalid, DTU will reset the power of GSM module, and then try first and second sets of IP/PORT again. This cycle continues until a successful connection.
- 1.8 Off-line data will be transmitted when the GPRS connection resumes.
- 1.9 All uploading data contain a GPRMC sentence and current status of DTU.
- 1.10 When status changes, DTU sends an immediate data.
- 1.11 Modify DTU's parameters by remote operation:
 - 1.11.1 GCP sends GTD command 5A_h to inquiry DTU's current parameter values.
 - 1.11.2 DTU sends DTG command 64_h to reply current parameters values.
 - 1.11.3 GCP sends DTG command 5B_h to set DTU's new parameter values.
 - 1.11.4 DTU sends DTG command 64_h to acknowledge the received new parameter values from GCP.
 - 1.11.5 If GCP confirms correctly two set of parameter values. It will send GTD command 6A_h to inform DTU to modify the parameter values.
 - 1.11.6 DTU updates parameter values.
 - 1.11.7 DTU sends DTG command 64_h to reply the result of parameter values updating, i.e. ERROR or OK.
 - 1.11.8 Finish modifying.

2 GTD Communication Protocol

2.1 Communication Packet

SYNC (2)	CID (2)	CH (1)	COMM (1)	SEQ (2)	PWK (4)	DATALEN (2)	DATA (0~1007)	EOT (3)
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2.2 Descriptions

2.2.1 SYNC(2 bytes)

Synchronization flag, always be C4_hD7_h

2.2.2 CID(2 bytes, LSB - MSB)

Caller identification, ranging from: 0000_h~FFFF_h

2.2.3 CH(1 byte)

01_h **GPRS (Default)**
02_h CDMA2000
03_h WCDMA

2.2.4 COMM(1 byte)

GCP to DTU command and description

00_h~0F_h System reserved
5A_h Inquire basic-parameter values of DTU
5B_h Set basic-parameter values of DTU
5C_h Inquire parameter values of GEO-FENCING
5D_h Set parameter values of GEO-FENCING
5E_h Inquire parameter values of SMS
5F_h Set parameter values of SMS
67_h Inquire GPS NMEA data
6A_h Execute the modification of parameter values
7B_h Enable one digital Output
7C_h Reset one digital Output

2.2.5 SEQ(2 bytes, LSB - MSB)

Serial number assigned to the command, ranging from 0000_h to FFFF_h. When exceeds FFFF_h, SEQ starts a new cycle from 0000_h

2.2.6 PWK(4 bytes, LSB - MSB)

$PWK = 25 \times UI(CID) + 333 \times UI(SEQ) + UI(ABCD_h)$

*UI(X): Unsigned Integer of X

2.2.7 DATALEN(2 bytes, LSB - MSB)

The value is the length of DATA in the communication packet.
0000_h is for no DATA, 000A_h is for 10 bytes DATA.

2.2.8 DATA(0~1007 bytes)

The DATA contents are different corresponding to each command (COMM)

5A _h	Inquire basic-parameter values of DTU	
Name	Length	Description
-	-	No DATA

5B _h	Set basic-parameter values of DTU	
Name	Length	Description
DTU_PARA	346	Please refer to 4.1 for details.

5C _h	Inquire parameter values of GEO-FENCING	
Name	Length	Description
-	-	No DATA

5D _h	Set parameter values of GEO-FENCING	
Name	Length	Description
GEO_PARA	256	Please refer to 4.2 for details.

5E _h	Inquire parameter values of SMS	
Name	Length	Description
-	-	No DATA

5F _h	Set parameter values of SMS	
Name	Length	Description
SMS_PARA	842	Please refer to 4.3 for details.

67 _h	Inquire GPS NMEA data	
Name	Length	Description
-	-	No DATA

6A _h	Execute the modification of parameter values	
Name	Length	Description
-	-	No DATA

7B _h	Enable one digital Output	
Name	Length	Description
OUTPUT_NUM	1	HEX type 01 _h → Enable 1 st OUTPUT 02 _h → Enable 2 nd OUTPUT

7C _h	Reset one digital Output	
Name	Length	Description
OUTPUT_NUM	1	HEX type 01 _h → Reset 1 st OUTPUT 02 _h → Reset 2 nd OUTPUT

2.2.9 EOT(3 bytes)

The end of the communication packet, it's always 04_h04_h04_h.

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3 DTG Communication Protocol

3.1 Communication Packet

SYNC (2)	CID (2)	CH (1)	COMM (1)	VAL (2)	STATUS (5)	IdleCnt (2)	IdleTime (2)
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Mileage (4)	TotalMileage (4)	GPRMC (80)	DATALEN (2)	DATA (0~914)	EOT (3)
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3.2 Descriptions

3.2.1 SYNC(2 bytes)

Synchronization flag, always be 3B_h28_h

3.2.2 CID(2 bytes, LSB - MSB)

Caller identification, ranging from: 0000_h ~ FFFF_h

3.2.3 CH(1 byte)

01_h DATA MODE
 02_h COMMAND MODE (for SLS-007CF)
 03_h COMMAND MODE (for SLS-007SF)
 05_h **COMMAND MODE (SLS-00886 default value)**

3.2.4 COMM(1 byte)

DTU to GCP command and description

00_h~0F_h System reserved
 18_h DTU's 1st data sent to GCP after power on
 19_h DTU's 1st data sent to GCP after connection failure
 62_h Immediate data uploading
 63_h Regular data uploading
 64_h Reply GCP commands
 70_h Send transparent data
 72_h Immediate data uploading with CELL LOCATE
 73_h Regular data uploading with CELL LOCATE

3.2.5 VAL(2 bytes, LSB - MSB)

Voltage values (mV), ranging from: 0000_h ~ EA60_h, 0VDC~60VDC
 2EE0_h is for 12V

3.2.6 STATUS(5 bytes)

5 Bytes, ordered as:

... VAL BYTE0 BYTE1 BYTE2 BYTE3 BYTE4 IdleCnt ...

BYTE0		
Name	BIT	Description
ACC_STAT	7	ACC, High Level 1, Low Level 0

GPS_MOD	6	GPS Module, normal 1, abnormal 0
GPS_ANT	5	GPS Antenna, normal 1, abnormal 0
TOW_ALERT	4	Movement / Towing alarm 1, in general 0
READ_SC	3	SIM card reading, normal 1, abnormal 0
SCH_DET	2	SIM card tray, inserted 1, not inserted 0
VOLT_ALERT	1	Voltage alarm 1, in general 0
GSM_MOD	0	GSM Module, normal 1, abnormal 0

BYTE1		
Name	BIT	Description
RESERVED	7	System reserved
GPRS_STAT	6	GPRS status, on-line 1, off-line 0
ACC_ALERT	5	ACC alarm 1, in general 0
RESERVED	4	System reserved
DIN1_NOW	3	1 st INPUT current level, High level 1, Low level 0
RESERVED	2	System reserved
DIN2_NOW	1	2 nd INPUT current level, High level 1, Low level 0
RESERVED	0	System reserved

BYTE2		
Name	BIT	Description
RESERVED	7	System reserved
DOUT1_NORM	6	1 st OUTPUT default level, High level 1, Low level 0
DOUT1_NOW	5	1 st OUTPUT current level, High level 1, Low level 0
DOUT2_NORM	4	2 nd OUTPUT default level, High level 1, Low level 0
DOUT2_NOW	3	2 nd OUTPUT current level, High level 1, Low level 0
RESERVED	2	System reserved
RESERVED	1	System reserved
RESERVED	0	System reserved

BYTE3		
Name	BIT	Description
RESERVED	7	System reserved
RESERVED	6	System reserved
RESERVED	5	System reserved
RESERVED	4	System reserved
RESERVED	3	System reserved
RESERVED	2	System reserved
RESERVED	1	System reserved
SECURITY	0	Security, enable 1, disable 0

BYTE4		
Name	BIT	Description
RESEND_DATA	7	Off-line data 1, current data 0
SPEED_ALERT	6	Speeding alarm 1, in general 0

G4_ALERT	5	4 th Geo-Fencing(Circle), alert 1, in general 0
G3_ALERT	4	3 rd Geo-Fencing(polygon), alert 1, in general 0
G2_ALERT	3	2 nd Geo-Fencing(polygon), alert 1, in general 0
G1_ALERT	2	1 st Geo-Fencing(polygon), alert 1, in general 0
BATT_STAT	1	Li-on battery voltage, normal 1, abnormal 0
VIN_STAT	0	Permanent battery voltage, normal 1, abnormal 0

3.2.7 IdleCnt(2 bytes, LSB - MSB)

Counting number of idling speeds, in general is 0000_h. Refresh the calculations when ACC is ON at each time.

3.2.8 IdleTime(2 bytes, LSB - MSB)

Idle in second, in general is 0000_h.

3.2.9 Mileage(4 bytes, LSB - MSB)

Moving distance, calculates the distance between current data, and latest previous data. Unit: meter

3.2.10 Total Mileage(4 bytes, LSB - MSB)

Total moving distance, refreshes the calculations when ACC is ON at each time. Unit: meter

3.2.11 GPRMC(80 bytes)

NMEA 0183 GPRMC data, contains the new line characters (0D_h0A_h), total length is 80 bytes. When the length of the GPS sentence is shorter than 80 bytes, please fill in 00_h for empty bytes

GPRMC format example:

\$GPRMC,hhmmss.sss,A,IIII.IIII,a,yyyyy.yyyy,a,x.xx,xxx.xx,ddmmyy,x.x,a*hh0D_h0A_h00_h00_h...
e.g. \$GPRMC,093843.000,A,2459.8205,N,12129.1458,E,0.04,143.69,270306,,*0F

3.2.12 DATALEN(2 bytes, LSB - MSB)

The value is the length of DATA in the communication packet.

0000_h is for no DATA, 000A_h is for 10 bytes DATA

3.2.13 DATA(0~914 bytes)

The total length of DATA contents should be equal to or less than defined 914bytes

18 _h /19 _h	DTU's 1 st data sent to GCP after power on / DTU's 1 st data sent to GCP after connection failure	
Name	Length	Description
PID	2	Product ID
VER	2	Version Number
FV	14	Version number of firmware

62 _h /63 _h	Immediate data uploading / Regular data uploading	
Name	Length	Description

-	-	No DATA
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64 _h	Reply GCP commands	
Name	Length	Description
RCMD	1	The command from GCP, please refer 2.2.4 for the detailed definition
RC	1	Result code 00 _h → ERROR 01 _h → OK 02 _h → Wait for check
RM	N	Data like DTU Parameters

70 _h	Send transparent data	
Name	Length	Description
PortNo	1	The source of the transparent data 00 _h → GPS module
TotalPktNo	2	HEX format, the total number of packets in 1 data ranges from 1~65535 (0001 _h ~FFFF _h)
PktNo	2	HEX format, sending order of total packets in 1 data ranges from 1~65535 (0001 _h ~FFFF _h)
PktDataLen	2	HEX format, the length of PktData ranges from 1~65535 (0001 _h ~FFFF _h)
PktData	1~N	Data contents, maximum 907 bytes

72 _h /73 _h	Immediate data uploading with CELL LOCATE / Regular data uploading with CELL LOCATE	
Name	Length	Description
CELL_LOC	N	Use nearby cellular towers to triangulate an approximate position CELL_LOC format example: UTC_DATE,UTC_TIME,LATITUDE,LONGITUDE e.g. 2011/12/23,07:17:06,24.994417,121.485191

3.2.14 EOT(3 bytes)

The end of the communication packet, it's always 04_h04_h04_h.

4 Parameter Descriptions

4.1 Basic-Parameters of DTU (346 bytes)

SN.	Parameter	Length	Type	Description
1	PID	2	Char	Product model ID SLS-00886 → G5
2	CID	2	HEX	Caller Identification
3	PWD	8	Char	Password
4	VER	2	Char	Version Number
5	ADT	1	HEX	Time to continue working after ACC is OFF (unit: minute) 00 _h → Immediate shutdown 01 _h ~ FE _h → 1~254 minutes FF _h → Never shut down
6	IP1	30	Char	First IP Address/Domain Name
7	PT1	5	Char	First Port
8	IP2	30	Char	Second IP Address/Domain Name
9	PT2	5	Char	Second Port
10	APN	40	Char	Access point name
11	USR	20	Char	GPRS service login account
12	PSS	20	Char	GPRS service login password
13	UT1	2	HEX	Uploading interval when ACC ON (unit: second) 0000 _h → Uploading OFF 0001 _h ~ FFFF _h → 1~65535s
14	UT2	2	HEX	Uploading interval when ACC OFF (unit: second) 0000 _h → Uploading OFF 0001 _h ~ FFFF _h → 1~65535s
15	THB	2	HEX	Heart beat uploading interval (unit: second) 0000 _h → Uploading OFF 0001 _h ~ FFFF _h → 1~65535s
16	ADC	2	HEX	Voltage values Setting (unit: mV) 0000 _h → Function OFF 0001 _h ~ EA60 _h → 1~60000 mV
17	CO1	3	HEX	1 st Digital Output Setting 【Byte0】 General level, Enabled setting 0x _h → Low level in general Fx _h → High level in general x0 _h → No action when enabled x1 _h ~ xE _h → Convert 1~14 times when enabled xF _h → Never stop converting when enabled 【Byte1】 Time to maintain the level

				00 _h ~ FE _h → Maintain 0~254s when enabled FF _h → Permanent converting to opposite level 【Byte2】 Time maintained to resume to general level 00 _h ~ FF _h → Resume to general level 0~255s
18	CO2	3	HEX	2 nd Digital Output Setting The setting is the same as CO1
19	OP1	20	Char	1 st Phone No. allowed dialing-in and get SMS
20	OP2	20	Char	2 nd Phone No. allowed dialing-in and get SMS
21	OP3	20	Char	3 rd Phone No. allowed dialing-in and get SMS
22	AN1	20	Char	1 st Phone No. allowed dialing-in
23	AN2	20	Char	2 nd Phone No. allowed dialing-in
24	AN3	20	Char	3 rd Phone No. allowed dialing-in
25	AN4	20	Char	4 th Phone No. allowed dialing-in
26	AN5	20	Char	5 th Phone No. allowed dialing-in
27	SPD	1	HEX	Speeding alarm (km/hr) 00 _h → Function OFF 01 _h ~ FF _h → 1~255 km/hr
28	IDS	1	HEX	The distance for idling (unit: meter) 00 _h → Function OFF 01 _h ~ FF _h → 1~255m
29	ITS	1	HEX	The time for idling (unit: 10 seconds) 00 _h → Function OFF 01 _h ~ FF _h → 10 to 2550s
30	FSL	2	HEX	【Byte0】 0: OFF 1: ON Bit0 → AGPS function Bit1 → GPRS auto-connection function Bit2 → Geo-Fencing function Bit3 → Idle detection function Bit4 → Auto remind security enabling SMS Bit5 → Off-line data transmission function Bit6 → Only send immediate off-line data function Bit7 → Only send latest 1 off-line data function 【Byte1】 0: OFF 1: ON Bit0 → Voltage values detection function Bit1 → high level trigger function Bit2 → low level trigger function Bit3 → First IP Address UDP connection function Bit4 → Second IP Address UDP connection

				function Bit5 → Power saving function Bit6 → NP850 protocol Bit7 → System reserved
31	PSL	1	HEX	0: Provide Internet services 1: Not provide Internet services Bit0 → Phone No. to Dial out / Send SMS Bit1 → 1 st Phone No. allowed dialing-in Bit2 → 2 nd Phone No. allowed dialing-in Bit3 → 3 rd Phone No. allowed dialing-in Bit4 → 4 th Phone No. allowed dialing-in Bit5 → 5 th Phone No. allowed dialing-in Bit6 → 6 th Phone No. allowed dialing-in Bit7 → 7 th Phone No. allowed dialing-in
32	LNG	1	HEX	The language in which to return Reverse Geocoding results 00 _h → Use the native language of the domain whenever possible 01 _h → ARABIC 02 _h → BASQUE 03 _h → BULGARIAN 04 _h → BENGALI 05 _h → CATALAN 06 _h → CZECH 07 _h → DANISH 08 _h → GERMAN 09 _h → GREEK 0A _h → ENGLISH 0B _h → ENGLISH (AUSTRALIAN) 0C _h → ENGLISH (GREAT BRITAIN) 0D _h → SPANISH 0E _h → BASQUE 0F _h → FARSI 10 _h → FINNISH 11 _h → FILIPINO 12 _h → FRENCH 13 _h → GALICIAN 14 _h → GUJARATI 15 _h → HINDI 16 _h → CROATIAN 17 _h → HUNGARIAN 18 _h → INDONESIAN 19 _h → ITALIAN 1A _h → HEBREW 1B _h → JAPANESE 1C _h → KANNADA 1D _h → KOREAN 1E _h → LITHUANIAN

				1F _h → LATVIAN 20 _h → MALAYALAM 21 _h → MARATHI 22 _h → DUTCH 23 _h → NORWEGIAN NYNORSK 24 _h → NORWEGIAN 25 _h → ORIYA 26 _h → POLISH 27 _h → PORTUGUESE 28 _h → PORTUGUESE (BRAZIL) 29 _h → PORTUGUESE (PORTUGAL) 2A _h → ROMANSCH 2B _h → ROMANIAN 2C _h → RUSSIAN 2D _h → SLOVAK 2E _h → SLOVENIAN 2F _h → SERBIAN 30 _h → SWEDISH 31 _h → TAGALOG 32 _h → TAMIL 33 _h → TELUGU 34 _h → THAI 35 _h → TURKISH 36 _h → UKRAINIAN 37 _h → VIETNAMESE 38 _h → CHINESE (SIMPLIFIED) 39 _h → CHINESE (TRADITIONAL)
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4.2 GEO-FENCING Parameters (256 bytes)

SN.	Parameter	Length	Type	Description
1	PS1	1	HEX	1 st Geo-Fencing setting (Polygon) x0 _h → Alert when outside the range x1 _h → Alert when inside the range nx _h → n coordinates FF _h → Function OFF
2	P1X1	4	HEX	1 st set: 1 st latitude (Floating point type) Datum → WGS84
3	P1Y1	4	HEX	1 st set: 1 st longitude (Floating point type) Datum → WGS84
4	P1X2	4	HEX	1 st set: 2 nd latitude (Floating point type) Datum → WGS84
5	P1Y2	4	HEX	1 st set: 2 nd longitude (Floating point type) Datum → WGS84
6	P1X3	4	HEX	1 st set: 3 rd latitude (Floating point type) Datum → WGS84
7	P1Y3	4	HEX	1 st set: 3 rd longitude (Floating point type) Datum → WGS84

8	P1X4	4	HEX	1 st set: 4 th latitude (Floating point type) Datum → WGS84
9	P1Y4	4	HEX	1 st set: 4 th longitude (Floating point type) Datum → WGS84
10	P1X5	4	HEX	1 st set: 5 th latitude (Floating point type) Datum → WGS84
11	P1Y5	4	HEX	1 st set: 5 th longitude (Floating point type) Datum → WGS84
12	P1X6	4	HEX	1 st set: 6 th latitude (Floating point type) Datum → WGS84
13	P1Y6	4	HEX	1 st set: 6 th longitude (Floating point type) Datum → WGS84
14	P1X7	4	HEX	1 st set: 7 th latitude (Floating point type) Datum → WGS84
15	P1Y7	4	HEX	1 st set: 7 th longitude (Floating point type) Datum → WGS84
16	P1X8	4	HEX	1 st set: 8 th latitude (Floating point type) Datum → WGS84
17	P1Y8	4	HEX	1 st set: 8 th longitude (Floating point type) Datum → WGS84
18	P1X9	4	HEX	1 st set: 9 th latitude (Floating point type) Datum → WGS84
19	P1Y9	4	HEX	1 st set: 9 th longitude (Floating point type) Datum → WGS84
20	P1X10	4	HEX	1 st set: 10 th latitude (Floating point type) Datum → WGS84
21	P1Y10	4	HEX	1 st set: 10 th longitude (Floating point type) Datum → WGS84
22	PS2	1	HEX	2 nd Geo-Fencing setting (Polygon) x0 _h → Alert when outside the range x1 _h → Alert when inside the range nx _h → n coordinates FF _h → Function OFF
23	P2X1	4	HEX	2 nd set: 1 st latitude (Floating point type) Datum → WGS84
24	P2Y1	4	HEX	2 nd set: 1 st longitude (Floating point type) Datum → WGS84
25	P2X2	4	HEX	2 nd set: 2 nd latitude (Floating point type) Datum → WGS84
26	P2Y2	4	HEX	2 nd set: 2 nd longitude (Floating point type) Datum → WGS84
27	P2X3	4	HEX	2 nd set: 3 rd latitude (Floating point type) Datum → WGS84
28	P2Y3	4	HEX	2 nd set: 3 rd longitude (Floating point type) Datum → WGS84
29	P2X4	4	HEX	2 nd set: 4 th latitude (Floating point type) Datum → WGS84

30	P2Y4	4	HEX	2 nd set: 4 th longitude (Floating point type) Datum → WGS84
31	P2X5	4	HEX	2 nd set: 5 th latitude (Floating point type) Datum → WGS84
32	P2Y5	4	HEX	2 nd set: 5 th longitude (Floating point type) Datum → WGS84
33	P2X6	4	HEX	2 nd set: 6 th latitude (Floating point type) Datum → WGS84
34	P2Y6	4	HEX	2 nd set: 6 th longitude (Floating point type) Datum → WGS84
35	P2X7	4	HEX	2 nd set: 7 th latitude (Floating point type) Datum → WGS84
36	P2Y7	4	HEX	2 nd set: 7 th longitude (Floating point type) Datum → WGS84
37	P2X8	4	HEX	2 nd set: 8 th latitude (Floating point type) Datum → WGS84
38	P2Y8	4	HEX	2 nd set: 8 th longitude (Floating point type) Datum → WGS84
39	P2X9	4	HEX	2 nd set: 9 th latitude (Floating point type) Datum → WGS84
40	P2Y9	4	HEX	2 nd set: 9 th longitude (Floating point type) Datum → WGS84
41	P2X10	4	HEX	2 nd set: 10 th latitude (Floating point type) Datum → WGS84
42	P2Y10	4	HEX	2 nd set: 10 th longitude (Floating point type) Datum → WGS84
43	PS3	1	HEX	3 rd Geo-Fencing setting (Polygon) x0 _h → Alert when outside the range x1 _h → Alert when inside the range nx _h → n coordinates FF _h → Function OFF
44	P3X1	4	HEX	3 rd set: 1 st latitude (Floating point type) Datum → WGS84
45	P3Y1	4	HEX	3 rd set: 1 st longitude (Floating point type) Datum → WGS84
46	P3X2	4	HEX	3 rd set: 2 nd latitude (Floating point type) Datum → WGS84
47	P3Y2	4	HEX	3 rd set: 2 nd longitude (Floating point type) Datum → WGS84
48	P3X3	4	HEX	3 rd set: 3 rd latitude (Floating point type) Datum → WGS84
49	P3Y3	4	HEX	3 rd set: 3 rd longitude (Floating point type) Datum → WGS84
50	P3X4	4	HEX	3 rd set: 4 th latitude (Floating point type) Datum → WGS84
51	P3Y4	4	HEX	3 rd set: 4 th longitude (Floating point type) Datum → WGS84

52	P3X5	4	HEX	3 rd set: 5 th latitude (Floating point type) Datum → WGS84
53	P3Y5	4	HEX	3 rd set: 5 th longitude (Floating point type) Datum → WGS84
54	P3X6	4	HEX	3 rd set: 6 th latitude (Floating point type) Datum → WGS84
55	P3Y6	4	HEX	3 rd set: 6 th longitude (Floating point type) Datum → WGS84
56	P3X7	4	HEX	3 rd set: 7 th latitude (Floating point type) Datum → WGS84
57	P3Y7	4	HEX	3 rd set: 7 th longitude (Floating point type) Datum → WGS84
58	P3X8	4	HEX	3 rd set: 8 th latitude (Floating point type) Datum → WGS84
59	P3Y8	4	HEX	3 rd set: 8 th longitude (Floating point type) Datum → WGS84
60	P3X9	4	HEX	3 rd set: 9 th latitude (Floating point type) Datum → WGS84
61	P3Y9	4	HEX	3 rd set: 9 th longitude (Floating point type) Datum → WGS84
62	P3X10	4	HEX	3 rd set: 10 th latitude (Floating point type) Datum → WGS84
63	P3Y10	4	HEX	3 rd set: 10 th longitude (Floating point type) Datum → WGS84
64	CS1	1	HEX	4 th Geo-Fencing setting (Circle) 10 _h → Alert when outside the range 11 _h → Alert when inside the range FF _h → Function OFF
65	C1X1	4	HEX	4 th set: Center latitude (Floating point type) Datum → WGS84
66	C1Y1	4	HEX	4 th set: Center longitude (Floating point type) Datum → WGS84
67	CR1	4	HEX	4 th set: Radius of the circle unit: KM (Floating point type, precise to 6 th decimal place)

4.3 SMS Parameters (842 bytes)

SN.	Parameter	Length	Type	Description
1	MS1	70	Char	1 st Digital Input triggered alert message
2	MS2	70	Char	2 nd Digital Input triggered alert message
3	MS3	70	Char	1 st Geo-Fencing triggered alert message
4	MS4	70	Char	2 nd Geo-Fencing triggered alert message
5	MS5	70	Char	3 rd Geo-Fencing triggered alert message
6	MS6	70	Char	4 th Geo-Fencing triggered alert message
7	MS7	70	Char	ACC ON abnormally alert message

8	MS8	70	Char	Abnormal permanent battery alert message
9	MS9	70	Char	Abnormal GPS antenna status alert message
10	MS10	70	Char	Abnormal voltage values alert message
11	MS11	70	Char	Speeding alert message
12	MS12	70	Char	Tow/Movement alert message
13	MSL	2	HEX	【Byte0】 0: OFF 1: ON Bit0 → Disable all messages Bit1 → Disable 1st Digital Input triggered alert message Bit2 → Disable 2nd Digital Input triggered alert message Bit3 → Disable 1st Geo-Fencing triggered alert message Bit4 → Disable 2nd Geo-Fencing triggered alert message Bit5 → Disable 3rd Geo-Fencing triggered alert message Bit6 → Disable 4th Geo-Fencing triggered alert message Bit7 → Disable ACC ON abnormally alert message 【Byte1】 0: OFF 1: ON Bit0 → Disable abnormal permanent battery alert message Bit1 → Disable abnormal GPS antenna status alert message Bit2 → Disable abnormal voltage values alert message Bit3 → Disable Speeding alert message Bit4 → System reserved Bit5 → System reserved Bit6 → System reserved Bit7 → System reserved

5 About RITI Technology Inc.

RITI Technology Inc., founded in 1992, is a leading technical service, software engineering and application service provider (ASP) company based in Taiwan. We specialize in the development and application of information technologies related to management, analysis, presentation and visualization of spatial and location-based information. Over the last 18 years, our technical expertise, along with our extensive connections with government, industry and academic institutions, has allowed us to establish a unique spatial database containing local digital data at the street level resolution and to grow into one of the leading spatial information companies in Taiwan.

Our expertise has allowed us to develop innovative spatial data products like EPMS (Electronic Patrol Management System), SuperLocator, DoVTs (Data over Voice Transmission System), and more. Meanwhile, we have provided customized professional solutions for clients in both the public and private realms.

With the unique digital spatial content and knowledge in GPS, GIS, wireless communication and Internet, we have also begun developing our business in internet ASP services, including a map-related information portal (www.map.com.tw) in 1998 and a web-based vehicle position portal (www.elocation.com.tw) in 2000.

Our business model:

